R13 pair-table prefetch gate: replacement-host run

Status, 2026-10-27: Five-round A/A and paired suite completed on EC2 instance `i-048baef2256adcc25`. The frozen holdout remains incomplete after two 60-attempt isolation stops on its final missing candidate row. The runner retained 267 holdout price records. This report records an incomplete run; it does not promote a Table A row.

The pre-timing protocol is `research/ic_tool_program/rounds/R13-prefetch-gate/PROTOCOL.md`. The candidate uses the admitted-run prefetch pass only for stored pair tables with at least 1,000,000 entries. Source revisions were base `261482646ee89b332dca69e2f296b0774c25d2c2` and candidate `efc6336d55c59cceb92cb82de73598ebe15ff3ca`. Both binaries were built on the run host with rustc 1.99.1 and the same Cargo.lock. The host was an Intel Xeon 6975P-C `m8i.8xlarge` in us-east-2b; isolation reserved both SMT siblings, CPUs 2 and 18, with one Rayon thread and unset `KIC_RHO_CANON`. The host manifest, runner configuration, raw records and refusal log are in the separate `runs-host-048baef2/` tree, copied to the R13 checkpoint store.

Before timing, native host-class checks passed, 24 frozen holdouts regenerated byte-for-byte, and 90/90 historical pin outputs matched. The A/A control completed 220 scheduled records and eight retained retry records with no missing cold pairs. At K0/F2^57, its base/copy cold ratio was 0.984873 with native 95% interval [0.976267, 0.993554]. The paired suite completed 400 scheduled records and 25 retries, again with no missing cold pairs.

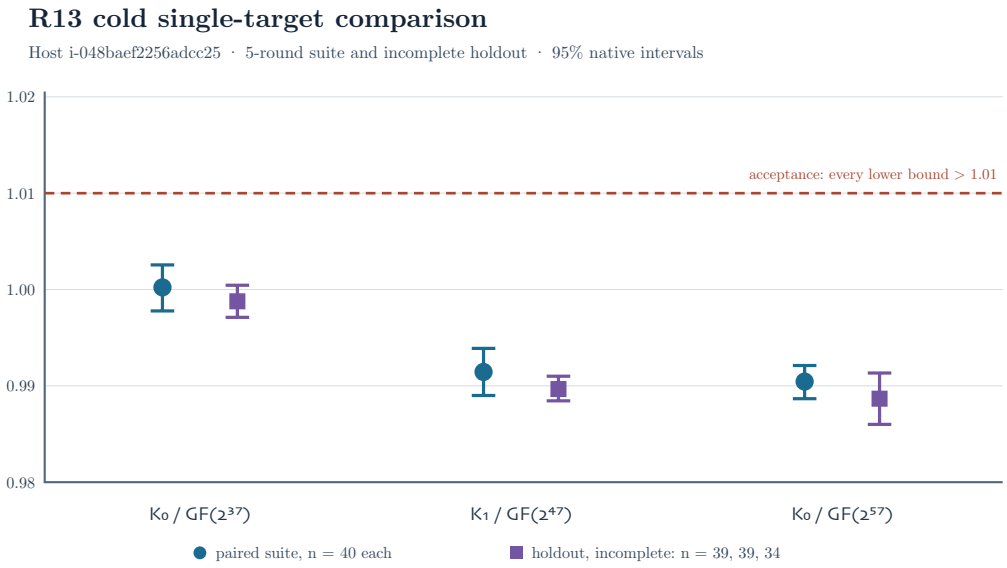


Figure 1: Cold base/candidate ratios at the three primary sizes. Native 95% intervals; suite n=40 at each size. Holdout counts 39, 39 and 34 are partial. Data: `graph-data.json`, from native R13 analysis of the retained run tree.

The suite's measured cold ratios were 1.000182 [0.997784, 1.002585] at K0/F2^37, 0.991425 [0.989017, 0.993839] at K1/F2^47, and 0.990408 [0.988664, 0.992154] at K0/F2^57. Values below one mean greater candidate cost in this cohort. The native suite analysis did not meet the preregistered lower-bound threshold above 1.01. Partial holdout ratios are plotted separately and remain provisional until every frozen pair is present.

The runner stopped twice on `cand/icv1-f2m57-tm747311035-c1f545af/M43-T177/r5.price.json` after 60 busy-machine refusals per attempt. The first stop was around 16:44 UTC; the same-host resume was started at 18:47 UTC and stopped at 19:13 UTC. Unrelated Git, EFS and research jobs used more CPU than the unchanged preflight's 0.20 CPU-second per two-second limit; another benchmark held the global lock during part of the wait. None of these attempts was admitted as a timed price record. The retained tree has 282 refusal lines. The incomplete schedule must be resumed only on this instance with the same binaries, CPU reservation and isolation guard after sustained quiet.

Independent native `ic_validate certverify` passed for the current host's 1,010 retained price records and 2,020 IC/rho scalar certificates. The former host's separate A/A tree was recovered read-only from snapshot `snap-0ac0539f266bbe1a6`: 220 scheduled records, 39 retries, 90 pin records and zero missing A/A cold pairs. Its K0/F2^57 ratio was 0.998332 [0.993237, 1.003453]. Its 349 records and 698 certificates also passed independent checks. Former-host timings are kept separate and do not fill current-host holdout pairs.

Remaining gates: complete the frozen holdouts; apply the prespecified interval-width extension rule; independently replay selected recorded runs and recompute all headline figures from committed raw data; attach a matched same-process rho reference and the 0/0.1/1 GAE band; render the final validated result and open a regular PR. The local release library test had 4,840 passes and nine inherited/environment failures, while the touched Koblitz filter passed 177 tests; the integration suite had one unchanged R07 analysis mismatch. These broad gates are not described as green.

The full descriptive Markdown report and paths to raw custody are in `REPORT.md` beside this PDF. No observation from this Intel CPU subclass is automatically a result for the Emerald Rapids Table A host class.

Evidence map and requirement status

- **Frozen identity — verified.** The same-host synthetic source snapshots were base `e0c1570149327ceae8c2a915f357acdd1dfe7e43` and candidate `574742e00deb26986b03ad4d2e4ac6e194843494`. The candidate source differs from the base only in `src/cryptanalysis/koblitz_index_calculus.rs`. Cargo.lock SHA-256 was `c2c0049f265e78432b79c5fe14a7ce6668cd44c3816012256dfcd7b40e5f4f6d`.
- **Executable identity — verified before timing and resume.** SHA-256: base IC `87d4dacf96cb369ec64038d9e8866ba2c44dbd4d51b890e0cbfa9651e6d054a5`; candidate IC `d0989e0273b44e6831942e080bc8208de23e37be7e8e8cb243ec53f3692769cf`; native runner `ca8b91924cfc540c86b1eb74e9ebf61b64eeeed6fa72e99a2c7fcb5c47bdfb6b`; isolation tool `03d5b835157361b997c9643b5320bb8df2a778463c9fe50d66bcad133b3ed3df`.
- **A/A and paired suite — complete on this host.** Native analysis found no missing cold pairs for 11 A/A sizes and 11 paired-suite sizes. Every timed record has an isolation receipt. A/A and suite outcomes are descriptive measurements on this specific Intel CPU subclass.
- **Frozen holdout — partial.** The runner retained 267 holdout price files. An isolation refusal is an infrastructure exclusion, not a timing value. Its two 60-attempt stops are in `holdout-new.log`, `holdout-resume.log` and `holdout/refusals.log`.
- **Independent checks — partial.** Native certificate checks passed for every retained IC and rho scalar certificate. Fresh binary replay and independent recomputation from repository raw data are still pending. The former host's separate A/A tree was verified and archived, not pooled.
- **Table A and benchmarks rows — not promoted.** The complete five-round holdout, matched host-class reproduction, full `VALIDATION.json`, and the repository's broad test gate remain open. No row is added to a leaderboard or benchmarks table by this partial report.

The current-host raw custody was uploaded under `s3://setec-astronomy-artifacts/checkpoints/crypto/codex__ic-single-target-tournament-20261026/20261027T184737Z-b89388529067/worker-runs-host-048baef2/`; its matching resume log is in `worker-logs-host-048baef2/`. The former-host archive is a separate object at checkpoint `20261027T165438Z-b89388529067/former-host-custody/r13-host-0250d75e-custody.tgz`, SHA-256 `aa765ef6fb37d8ec8e2604e6544edf885d445d2c796b4324bd8dd47edff34a7b`. The original EBS volume and completed snapshot remain available independently of the temporary recovery clone.